

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000486 - Models And Methods For Process Improvement And Assessments

DEGREE PROGRAMME

10AM - Master Universitario En Ingenieria Del Software

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	103000486 - Models And Methods For Process Improvement And Assessments
No of credits	4 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	10AM - Master Universitario en Ingeniería del Software
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Jose Antonio Calvo-Manzano Villalon (Subject coordinator)	5107	joseantonio.calvomanzano@upm.es	M - 10:30 - 13:30 W - 10:30 - 13:30
Tomas San Feliu Gilabert	5107	tomas.sanfeliu@upm.es	M - 10:00 - 13:00 W - 10:00 - 13:00

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Knowledge about the software life cycle processes (activities and tasks)

4. Skills and learning outcomes *

4.1. Skills to be learned

CE10 - Evaluar de forma objetiva los procesos y productos frente a los estándares y normas aplicables.

CE9 - Definir, evaluar y mejorar los procesos software de una organización.

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio (RD)

CG17 - Habilidades de gestión y capacidad de liderar un equipo que puede estar integrado por disciplinas y niveles distintos

CG18 - Capacidad de trabajar y comunicarse también en contextos internacionales

CG3 - Que los estudiantes sepan comunicar sus conclusiones y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades (RD)

4.2. Learning outcomes

RA36 - Posee dotes para liderar el cambio dentro de la organización

RA37 - Está capacitado para introducir mejoras en la organización

RA35 - Es capaz de definir, evaluar y mejorar los procesos software de una organización

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

New technologies are changing our way of life: airplanes, trains, communications, e-commerce, etc, and all of them need software to run. However, software is developed with poor quality; overruns in resources, effort and budget; and delays in delivery. It means that organizations must improve their way of producing software. In this context, it appears software process improvement. It consists of applying consistently the practices that provide good results, and changing those that cause problems.

So, enterprises should know their capacity/maturity level in order to develop software. The reference models are CMMI (Capability Maturity Model Integration with the views of Development, Services and Supplier Management), ISO 15504, and ITIL (Information Technology Infrastructure Library) among others. In order to be involved in a process improvement initiative, organizations should follow a process improvement lifecycle.

In this subject, and independently of the reference model used, a generic process improvement lifecycle oriented to large enterprises as well as small and medium-sized enterprises will be explained. In the main phase (the assessment phase), a method to make an assessment will be described. Students will have to make a plan in order to know the estimated hours and costs, resources and schedule involved in a process improvement initiative.

In the same way, a process improvement initiative in a small enterprise will be presented to students and they will have to analyse if the initiative is going to get benefits.

Later, the generic information related to the most known reference model (i.e., CMMI) will be explained

(maturity/capability levels, and institutionalization).

Finally, students will have to assess their estimating process and make a brief analysis and presentation.

5.2. Syllabus

1. Introduction

- 1.1. Maturity Profile
- 1.2. State of the practice
- 1.3. Process Improvement (PI) concepts & Justifying PI

2. PI life cycles

- 2.1. IDEAL, ISO 15504, Action Focus Improvement Model (AFIM)
- 2.2. AFIM: Commitment
- 2.3. AFIM: Assessment
- 2.4. AFIM: Infrastructure&Action Plans, and Implementation
- 2.5. A PI initiative in a small company

3. Reference Process Models

- 3.1. CMMI Model
- 3.2. The Estimating Process

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Chapter 1: Maturity Profile Report Duration: 00:30 Lecture Chapter 1: State of the practice. Process Improvement (PI) concepts. Duration: 01:30 Lecture			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
2	Chapter 1: Justifying PI Duration: 00:30 Lecture Chapter 2: Process Improvement life cycles Duration: 00:30 Lecture Chapter 2: Process Improvement life cycles: AFIM: Commitment Duration: 00:30 Lecture Exercise related to ROI in Commitment Duration: 00:30 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
3	Chapter 2: Process Improvement life cycles: AFIM: Commitment Duration: 01:30 Lecture Exercise related to ROI in Commitment Duration: 00:30 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
4	Exercise related to ROI in Commitment Duration: 02:00 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00 Exercise related to ROI in Commitment Other assessment Progressive assessment Presential Duration: 00:10 Exercise related to ROI in Commitment: Best Presentation Award Other assessment Progressive assessment

				Presential Duration: 00:05
5	Chapter 2: Process Improvement life cycles: AFIM: Assessment Duration: 01:30 Lecture Exercise related to ROI in Assessment Duration: 00:30 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
6	Chapter 2: Process Improvement life cycles: AFIM: Assessment Duration: 01:30 Lecture Exercise related to ROI in Assessment Duration: 00:30 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
7	Chapter 2: Process Improvement life cycles: AFIM: Assessment Duration: 01:30 Lecture Exercise related to ROI in Assessment Duration: 00:30 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
8	Exercise related to ROI in Assessment Duration: 02:00 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00 Exercise related to ROI in Assessment Group work Progressive assessment Presential Duration: 00:15 Exercise related to ROI in Assessment: Best Presentation Award Other assessment Progressive assessment Presential Duration: 00:05
9	Chapter 2: Process Improvement life cycles: AFIM: Action Plans and Implementation Duration: 01:00 Lecture Exercise related to ROI in a Small Enterprise (statement) Duration: 01:00 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00

10	Exercise related to ROI in a Small Enterprise Duration: 02:00 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00 Exercise related to ROI in a Small Enterprise Group work Progressive assessment Presential Duration: 00:10 Exercise related to ROI in a Small Enterprise: Best Presentation Award Other assessment Progressive assessment Presential Duration: 00:05
11	Chapter 3: CMMI Duration: 02:00 Lecture			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
12	Chapter 3: CMMI Duration: 01:00 Lecture Chapter 3: CMMI: Estimating process Duration: 01:00 Lecture			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00
13	Exercise related to Assessment: the Estimating process Duration: 02:00 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00 Exercise related to Assessment: the Estimating process Group work Progressive assessment Presential Duration: 00:10 Exercise related to Assessment: in the Estimating process: Best Presentation Award Group work Progressive assessment Presential Duration: 00:05
14	Progressive Evaluation: Review Duration: 02:00 Lecture			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00

15	Progressive Evaluation Duration: 02:00 Problem-solving class			Discussion/Participating in class (NON-RECOVERABLE) Other assessment Progressive assessment Presential Duration: 02:00 Progressive Evaluation Written test Progressive assessment Presential Duration: 02:00
16				Global Evaluation Written test Global examination Presential Duration: 02:00
17				

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Discussion/Participating in class (NON- RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
2	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
3	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
4	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
4	Exercise related to ROI in Commitment	Other assessment	Face-to-face	00:10	10%	5 / 10	CE9 CE10 CG17 CG18 CG1 CG3
4	Exercise related to ROI in Commitment: Best Presentation Award	Other assessment	Face-to-face	00:05	0%	0 / 10	CG18 CG3
5	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
6	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
7	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	5 / 10	CG18 CG1
8	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
8	Exercise related to ROI in Assessment	Group work	Face-to-face	00:15	20%	5 / 10	CE9 CE10 CG17 CG18 CG1 CG3
8	Exercise related to ROI in Assessment: Best Presentation Award	Other assessment	Face-to-face	00:05	%	0 / 10	CG18 CG3

9	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
10	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
10	Exercise related to ROI in a Small Enterprise	Group work	Face-to-face	00:10	15%	5 / 10	CE9 CE10 CG17 CG18 CG1 CG3
10	Exercise related to ROI in a Small Enterprise: Best Presentation Award	Other assessment	Face-to-face	00:05	%	0 / 10	CG18 CG3
11	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
12	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
13	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
13	Exercise related to Assessment: the Estimating process	Group work	Face-to-face	00:10	15%	5 / 10	CE9 CE10 CG17 CG18 CG1 CG3
13	Exercise related to Assessment: in the Estimating process: Best Presentation Award	Group work	Face-to-face	00:05	%	0 / 10	CG18 CG3
14	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
15	Discussion/Participating in class (NON-RECOVERABLE)	Other assessment	Face-to-face	02:00	1%	0 / 10	CG18 CG1
15	Progressive Evaluation	Written test	Face-to-face	02:00	25%	5 / 10	CE9 CE10 CG18 CG1 CG3

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
16	Global Evaluation	Written test	Face-to-face	02:00	85%	5 / 10	CE9 CE10 CG17 CG18 CG1 CG3

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Final Test	Written test	Face-to-face	03:00	100%	5 / 10	CE9 CE10 CG17 CG18 CG1 CG3

7.2. Assessment criteria

The assessment activities are the following:

1. Attendance and Active participation of students during the classes (15%). This is a NON-RECOVERABLE activity because the specific class and its discussion is not going to be repeated. This is an INDIVIDUAL activity.
2. Group Exercises related to Return on Investment (ROI) in Commitment (10%), Assessment (20%).. These are TEAM activities.
3. Group Exercise related to ROI in a Small enterprise (15%). This is a TEAM activity
4. Group Exercise related to the Assessment of the Estimating Process (15%). This is a TEAM activity.
5. Individual Progressive evaluation exam (25%) related to the main concepts of the subject. This is an INDIVIDUAL activity.

It should be noted that to motivate students in making presentations, there are some Best Presentation Award activities. The way to choose the best presentation is by voting with the rest of the students (but you cannot vote for the team to which you belong to). The team that wins the best presentation award will have 0.1 extra points in the final grade for the subject.

The previous assessment activities can be recovered (except activity 1), if failed, in the Global Evaluation Test (16 week). In order to recover the activities, the student must have obtained a grade greater than zero in the corresponding activity.

Students should have a rating greater or equal than 5.0 (over 10) to pass each activity (except activity 1). If not, the activities can be recovered in the Global Evaluation Test (when grading is greater than zero)..

In the global evaluation (week 16), the grade of this exam computes the 85% of the final grade. The remaining 15% corresponds to the rating obtained in the previous activity 1.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Subject Moodle Site	Web resource	Students will be able to get the slides and other resources from the Moodle platform.
IDEAL	Bibliography	McFeeley, B.; IDEAL: A User's Guide for Software Process Improvement; Handbook CMU/SEI-96-HB-001; February 1996
SCAMPI	Bibliography	SCAMPI Upgrade Team; Standard CMMI® Appraisal Method for Process Improvement (SCAMPI) A, Version 1.3: Method Definition Document; Handbook CMU/SEI 2011-HB-?001; March 2011
CMMI Model V2.0	Bibliography	CMMI Institute, CMMI V2.1 Model, December 2018

9. Other information

9.1. Other information about the subject

This course implements various innovative teaching methodologies to motivate and reinforce student learning.

1) The Virtual Collaboration methodology is implemented in all team exercises. Students, grouped into teams, can solve problems in person or collaborate through ICT (using applications such as Teams or Zoom for communication and OneDrive for file sharing). This allows students to connect remotely and work as a team. Furthermore, the instructor can join one of these meetings if necessary to address any questions or concerns.

2) The Learning-by-Doing methodology is applied in the course by defining the problem to be solved, which is developed in two or three exercises. What students do in the first exercise serves as a guide for what they will do in the second (and, if applicable, a third). To this end, the instructor initially explains the theoretical aspects of the different activities along with a practical example, and then the students, in teams of three to five members, carry out these activities. The teacher evaluates the first exercise and provides feedback to the students so that subsequent exercises can take it into account.